LEARNER GUIDE

Parallel stories of change

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

You can now write a line as y = mx + c. This lesson looks at two lines at once, and asks what makes them parallel — running side by side and never meeting. The answer is all about the gradient: parallel lines change at exactly the same rate (the same m), but they can begin in different places (different c). You will pick out parallel lines, build one yourself by matching a gradient, decide whether pairs of lines are parallel or not, and compare two real plans that charge the same rate. The question you are exploring is: can two lines change alike but begin differently?

What you are learning to notice

- That parallel lines never meet — they stay the same distance apart.
- That parallel lines have the same gradient m: they are exactly as steep as each other.
- That their intercepts c are usually different — that is what stacks one above the other.
- That crossing the y-axis at the same point does NOT make lines parallel; only equal gradients do.

Moving through it, one step at a time

In the Parallel Line Challenge, look at a line and pick the equation with the same gradient — that is the parallel one. In Make it parallel, set your gradient to match the reference and choose any different intercept; it turns green when the lines are genuinely parallel. In Parallel or not?, compare two equations by their gradients alone. Then compare two real plans that charge the same rate: write both equations, see they are parallel, and work out which is always cheaper. There is no rush; use the coordinate grid on the Scratchpad to draw two lines with the same gradient and check they never meet.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (reading the gradient and the intercept from an equation, and knowing that lines are equally steep when their gradients match), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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